

Day 1 — Introduction to Generative AI, GPT, and Modern AI Systems

Course

GPT + AI Agents + RAG + n8n + API Integration + Business Automation

Lecture Duration

60 Minutes

Level

Beginner

Day 1 Main Goal

By the end of this lecture, students should understand:

- What Artificial Intelligence is
- What Generative AI is
- What an LLM is
- What GPT means
- How GPT generates an answer
- What tokens are
- What context is
- The difference between training and inference
- What AI can and cannot do
- What hallucination means
- How businesses use generative AI
- Where GPT fits into AI Agents, RAG, APIs, n8n, and business automation
- Why AI should be treated as a component of a system rather than magic

1. 60-Minute Lecture Schedule

Time	Section
0–5 min	Welcome and course introduction
5–12 min	What is Artificial Intelligence?

Time	Section
12–20 min	Traditional AI vs Generative AI
20–28 min	What are LLMs and GPT?
28–36 min	How GPT works
36–42 min	Tokens, prompts, context, and output
42–48 min	Training vs inference
48–53 min	AI capabilities and limitations
53–57 min	Business use cases
57–60 min	Exercise, quiz, recap, homework

2. Opening — 0 to 5 Minutes

Instructor Introduction

Start by asking students:

How many of you have used ChatGPT, Claude, Gemini, Copilot, or another AI assistant?

Then ask:

When you type something into ChatGPT, what do you think is actually happening?

Typical beginner answers may include:

- It searches Google.
- It knows everything.
- It looks information up from a database.
- It understands what I mean.
- It thinks like a human.
- It copies information from websites.

Use these answers to introduce the main topic.

Instructor Explanation

Say:

Artificial Intelligence is much broader than ChatGPT.

ChatGPT is one application of a category called **Generative AI**.

Generative AI systems can create new outputs such as:

- Text
- Images
- Audio
- Video
- Code
- Data
- Summaries
- Analysis
- Recommendations

During this course, we will focus mainly on AI systems that work with language and business information.

Our final goal is not simply:

"How do I use ChatGPT?"

Our final goal is:

"How do I design an AI system that can understand information, access business knowledge, use software tools, make decisions, and automate work?"

Show the progression:

```
graph TD; A[ChatGPT User] --> B[Prompt Engineering]; B --> C[GPT API]; C --> D[Structured AI]; D --> E[RAG]; E --> F[AI Agents]; F --> G[API Integration]; G --> H[n8n Automation]; H --> I[Business AI System];
```

ChatGPT User
↓
Prompt Engineering
↓
GPT API
↓
Structured AI
↓
RAG
↓
AI Agents
↓
API Integration
↓
n8n Automation
↓
Business AI System

Tell students:

Today we are learning the foundation underneath everything else.

3. What Is Artificial Intelligence? — 5 to 12 Minutes

Simple Definition

Artificial Intelligence is the broad field of creating computer systems capable of performing tasks that normally require some form of human intelligence.

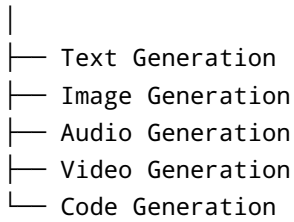
Examples include:

- Recognizing images
 - Understanding speech
 - Translating languages
 - Detecting fraud
 - Predicting customer behavior
 - Recommending products
 - Playing games
 - Generating text
 - Writing code
 - Answering questions
-

4. AI Is an Umbrella Term

Explain the hierarchy:

```
Artificial Intelligence
|
├── Machine Learning
|   |
|   ├── Deep Learning
|   |
|   ├── Computer Vision
|   |
|   ├── Speech AI
|   |
|   ├── Robotics
|   |
|   └── Generative AI
```

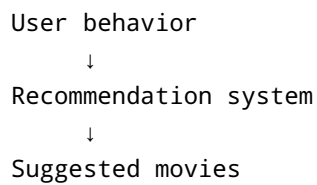


Explain that these categories can overlap.

5. Everyday Examples of AI

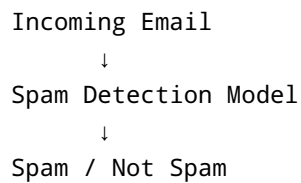
Ask students whether these systems contain AI.

Netflix recommendations

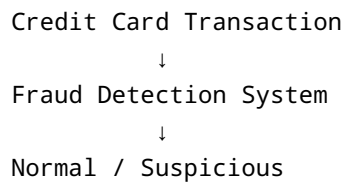


AI can analyze previous behavior and estimate which content the user may enjoy.

Email spam detection



Fraud detection



Facial recognition

```
Image
↓
Computer Vision Model
↓
Face Detection / Identification
```

ChatGPT

```
User Prompt
↓
Language Model
↓
Generated Response
```

Highlight:

Not all AI is generative AI.

6. Traditional Software vs AI

This distinction is important.

Traditional Software

A programmer writes explicit rules.

Example:

```
if age >= 18:
    print("Adult")
else:
    print("Minor")
```

The developer defines the logic.

Input:

```
Age = 25
```

Program:

```
IF age >= 18
```

Output:

```
Adult
```

7. AI-Based System

Suppose we want to classify customer messages.

Customer says:

I have emailed twice about my broken product and nobody has helped me.

Writing thousands of rules would be difficult.

Instead an AI model could produce:

```
{  
  "category": "product_issue",  
  "sentiment": "negative",  
  "priority": "high"  
}
```

The AI learned patterns from data rather than requiring a programmer to write a rule for every possible sentence.

8. Important Mental Model

Traditional programming often looks like:

```
Rules + Data
```

```
↓
```

Software
↓
Answer

Machine learning is closer to:

Examples + Learning
↓
Model
↓
New Input
↓
Prediction

Generative AI adds generation:

Prompt
↓
Generative Model
↓
New Content

9. What Is Generative AI? — 12 to 20 Minutes

Definition

Generative AI is AI designed to create new content based on patterns learned from data.

Examples:

Text

Prompt:

Write a professional email thanking a customer.

Output:

A newly generated email.

Image

Prompt:

Generate an image of a futuristic office.

Output:

A newly generated image.

Code

Prompt:

Write a Python function that sorts customer records by date.

Output:

New Python code.

Audio

AI can generate:

- Speech
- Voices
- Sound effects
- Music

Video

AI can generate or transform video content from text, images, or other media.

10. Traditional AI vs Generative AI

Use this table.

Traditional AI	Generative AI
Predicts	Generates
Classifies	Creates
Detects	Writes
Scores	Summarizes

Traditional AI	Generative AI
Recommends	Explains
Identifies	Produces new content

Important clarification:

Generative AI can also classify, analyze, extract, and predict in some workflows.

The distinction is about the model's ability to **generate output**, not a strict boundary between every use case.

11. Classification Example

Input:

I want my money back.

Traditional classifier:

Refund

Generative AI could return:

```
{
  "intent": "refund",
  "sentiment": "negative",
  "urgency": "medium",
  "recommended_action": "verify order and refund eligibility"
}
```

It can perform multiple language tasks simultaneously.

12. Why Generative AI Became Important

Explain five reasons.

1. Natural Language Interface

People can communicate with software using ordinary language.

Instead of:

```
SELECT *  
FROM orders  
WHERE customer_id=10034;
```

a user could ask:

Show me John Smith's latest order.

2. General-Purpose Capability

One model may perform many tasks:

```
Summarize  
Translate  
Classify  
Extract  
Generate  
Explain  
Rewrite  
Analyze  
Code  
Plan
```

Previously, many tasks required separate systems.

3. Easy Integration

Developers can integrate language models into:

- Websites
- Mobile applications
- CRMs
- Ecommerce platforms
- Help desks
- Internal tools

- Workflow systems
 - n8n
 - Custom applications
-

4. Automation Potential

Generative AI can handle unstructured information.

Businesses contain enormous amounts of:

- Emails
- Documents
- Customer requests
- Meeting transcripts
- Product descriptions
- Support tickets
- Reports

AI can help convert this information into structured actions.

5. Human-Friendly Interaction

Instead of teaching employees complex software commands, AI can act as a natural-language interface.

Example:

Give me a summary of today's high-priority customer complaints.

13. What Is an LLM? — 20 to 28 Minutes

LLM = Large Language Model

Break the phrase apart.

Large

The system is trained using a very large amount of data and contains a very large number of learned parameters.

Language

It primarily works with language representations.

Model

A model is a learned mathematical system that maps inputs to outputs.

So:

A Large Language Model is a learned AI model designed to process and generate language.

14. Examples of LLM Tasks

An LLM can perform:

```
Question Answering
Summarization
Translation
Classification
Extraction
Writing
Brainstorming
Reasoning
Coding
Document Analysis
Information Transformation
```

15. What Does GPT Mean?

GPT stands for:

G — Generative

It generates new output.

P — Pre-trained

It learns broad language patterns before you start interacting with it.

T — Transformer

Transformer refers to the neural-network architecture behind the model family.

For Day 1, students do not need the mathematics of transformers.

Use this simple definition:

GPT is a type of large language model designed to generate and work with language.

16. GPT Is Not the Same as ChatGPT

This is an important beginner distinction.

GPT

A model.

ChatGPT

An application that gives users an interface and additional capabilities around AI models.

Simple analogy:

Engine → GPT Model

Car → ChatGPT Application

The engine is an important component, but the complete product contains much more.

17. Model vs Application vs AI System

Teach three layers.

Layer 1 — Model

GPT

Receives input and generates output.

Layer 2 — Application

```
Chat Interface
  ↓
  GPT
```

Provides user experience around the model.

Layer 3 — AI System

```
User
  ↓
Application
  ↓
AI Agent
├─ GPT
├─ Company Knowledge
├─ APIs
├─ Database
└─ Business Tools
```

This third level is where the course will eventually go.

18. How Does GPT Work? — 28 to 36 Minutes

Do not introduce heavy mathematics.

Use a simple sentence.

Input:

The sky is usually __

The model may estimate possibilities like:

blue	85%
clear	5%
cloudy	4%
bright	3%
other	3%

It selects a likely continuation.

Then it continues:

```
The
↓
The sky
↓
The sky is
↓
The sky is usually
↓
The sky is usually blue
```

Explain:

GPT generates output step by step.

At each step it estimates what token should come next based on:

- The current input
- Previous tokens
- Patterns learned during training

19. The Core Idea

Use this simplified diagram:

```
Input Text
↓
Convert to Tokens
↓
Model Processes Context
↓
Predict Next Token
↓
Add Token
↓
Predict Next Token
↓
Continue
```

↓
Final Response

Tell students:

GPT is fundamentally a prediction system, but because its learned language patterns are extremely rich, next-token prediction can produce surprisingly sophisticated behaviors.

20. Does GPT Think Like a Human?

For beginners, avoid claiming either extreme.

Say:

GPT can perform tasks that appear reasoning-like, but its internal operation is different from human thought.

It does not automatically have:

- Human experiences
- Human emotions
- Personal beliefs
- Human consciousness
- Direct access to reality
- Perfect understanding

For engineering purposes, the important rule is:

Evaluate the system by what it can reliably do, not by assuming human-like mental states.

21. What Is a Token? — 36 to 42 Minutes

GPT does not process language exactly as humans see words.

It processes units called **tokens**.

A token might represent:

- A complete word
- Part of a word
- Punctuation
- Other text fragments

Conceptual example:

```
"Artificial intelligence is powerful."
```

could be represented approximately as:

```
Artificial  
intelligence  
is  
powerful  
.
```

But actual tokenization can differ.

22. Why Tokens Matter

Tokens affect:

- How much information a model can receive
- Response length
- Processing cost in API systems
- Context limits
- Performance

Explain:

When using an AI API, usage is often measured partly in tokens.

Simple conceptual formula:

```
Input Tokens  
+  
Output Tokens  
=  
Total Token Usage
```

23. What Is a Prompt?

A **prompt** is the information or instruction given to the AI.

Example:

Summarize this document in five bullet points.

That is a prompt.

A richer prompt might contain:

```
Role
Context
Task
Rules
Examples
Output Format
```

Example:

```
You are a customer-support assistant.

Analyze the following customer message.

Return:
- Summary
- Intent
- Sentiment
- Priority

Customer message:
"I ordered my product 10 days ago and it hasn't arrived."
```

Students will learn prompt engineering later in the course.

24. What Is Context?

Context is the information available to the model while generating the current response.

Example conversation:

```
User:
My company is called ABC Shoes.

AI:
```

Understood.

User:
Write a slogan for my company.

The model can use the earlier information:

ABC Shoes

because it is part of the conversation context.

25. Context Window

A model cannot necessarily process unlimited information at once.

The maximum amount of information it can consider is often called its **context window**.

Conceptually:

```
Context Window
|
├─ Instructions
├─ Conversation
├─ Documents
├─ Tool Results
├─ User Question
└─ Previous Outputs
```

Everything consumes context.

26. Why Context Matters for Business AI

Imagine a customer-support agent receives:

```
Company Policy
+
Customer Profile
+
Order Information
```

+
Conversation History
+
Current Question

The AI needs enough relevant context to answer accurately.

This becomes especially important later when we study **RAG**.

27. Training vs Inference — 42 to 48 Minutes

Beginners often confuse these.

Training

During training, the AI model learns patterns from large amounts of data.

Simplified:

```
Large Dataset
  ↓
Training Process
  ↓
Model Learns Patterns
  ↓
Trained Model
```

Training is computationally expensive.

28. Inference

Inference happens when we use the trained model.

```
Trained Model
  +
User Prompt
  ↓
Inference
```

↓
Response

Every normal ChatGPT interaction involves model inference.

29. Simple Analogy

Use a student analogy.

Training

A student studies:

- Books
- Lessons
- Exercises
- Examples

Inference

During an exam, the student receives a new question and produces an answer based on what was learned.

The analogy is imperfect, but useful for beginners.

30. Important Difference: Model Knowledge vs Current Information

Explain:

The information learned during training is not necessarily the same as information available right now.

A model may need additional tools for:

- Today's weather
- Current prices
- Live inventory
- Recent emails
- Today's calendar
- Latest stock data
- New company policies

This leads to a critical course concept:

GPT Alone
↓
Limited to available model/context information

GPT + Tools
↓
Can access external systems

Later students will learn to connect GPT to tools using:

- APIs
- AI Agents
- n8n
- Databases
- Search
- RAG

31. What GPT Is Good At — 48 to 53 Minutes

GPT is especially useful for language-heavy work.

1. Summarization

Input:

Long document.

Output:

Short summary.

2. Classification

Input:

I haven't received my order.

Output:

```
{  
  "category": "shipping_issue"  
}
```

3. Extraction

Input:

My name is Sarah and I want 15 units delivered to Denver.

Output:

```
{  
  "name": "Sarah",  
  "quantity": 15,  
  "location": "Denver"  
}
```

4. Transformation

Input:

Informal email.

Output:

Professional email.

5. Generation

GPT can generate:

- Emails
- Reports
- Product descriptions
- Documentation
- Scripts
- Code
- Marketing copy

6. Analysis

Input:

Customer feedback.

Output:

- Common complaints
- Trends
- Sentiment
- Possible actions

7. Question Answering

GPT can answer questions based on:

- Existing model knowledge
- Information supplied in the prompt
- Retrieved documents
- External tools

32. What GPT Is Not Automatically Good At

This section is extremely important.

GPT Does Not Automatically Know Live Data

Question:

How many products are currently in our warehouse?

GPT cannot know unless it has access to inventory data.

Correct architecture:

```
GPT
↓
Inventory API
↓
```

Live Database
↓
Current Stock

33. GPT Does Not Automatically Know Private Company Data

Suppose your company's return policy changed yesterday.

The model does not magically know it.

Correct architecture:

Company Documents
↓
RAG
↓
GPT

34. GPT Does Not Automatically Perform Actions

A normal language model does not automatically:

- Send an email
- Refund money
- Modify a CRM
- Create an invoice
- Schedule a meeting

To perform actions, it needs tools.

GPT
↓
Tool
↓
Business System

That leads directly to **AI Agents**.

35. What Is AI Hallucination?

A hallucination occurs when an AI generates information that appears plausible but is unsupported or incorrect.

Example:

User:

What is the refund period in our company policy?

AI:

Customers can request a refund within 45 days.

But suppose the actual policy says:

30 days.

The AI generated believable but incorrect information.

That is a serious problem.

36. Why Hallucinations Happen

Remember:

GPT is generating likely language.

It is not automatically checking every statement against a trusted database.

Therefore:

Fluent
≠
Correct

and:

Confident
≠
Correct

This is one of the most important AI principles students should remember.

37. How We Reduce Hallucination

Introduce techniques students will learn later:

```
Better Instructions
      +
Structured Output
      +
RAG
      +
Tools
      +
Validation
      +
Human Approval
      =
More Reliable AI System
```

38. Hallucination Example

Ask:

What is the annual leave policy at ABC Corporation?

If no company document is provided, GPT should ideally say:

I don't have enough information to determine ABC Corporation's leave policy.

A poorly designed system might invent:

Employees receive 25 days per year.

Explain:

A good AI system should know when it lacks evidence.

39. Other AI Limitations

Discuss briefly.

Accuracy

AI can make mistakes.

Bias

Models may reflect patterns or biases in training data or instructions.

Security

Sensitive data requires careful handling.

Privacy

Companies should determine which information can be sent to external AI services.

Consistency

The same prompt may sometimes produce different outputs.

Context

Important information may be missing.

Cost

Large-scale AI processing can cost money.

Latency

AI responses are not instantaneous in every system.

40. Human-in-the-Loop

Introduce this concept early.

Some decisions should require human approval.

Example:

```
Customer Refund Request
    ↓
    AI
    ↓
Refund = $10?
    ↓
Auto Process

Refund = $10,000?
    ↓
Human Approval
```

Tell students:

The best business automation is not necessarily 100% automation.

Sometimes the best design is:

```
AI handles routine work.
Humans handle important exceptions.
```

41. Business Use Cases — 53 to 57 Minutes

Now connect everything to real businesses.

Customer Support

```
Customer Message
    ↓
    GPT
    ↓
Classify
```

Summarize
Draft Response

Later:

Customer
↓
AI Agent
├─ RAG
├─ Order API
└─ CRM
↓
Response

42. Sales

New Lead
↓
GPT Analysis
↓
Lead Score
↓
Personalized Email
↓
CRM

43. Marketing

GPT can help with:

- Content ideas
 - Social posts
 - Product descriptions
 - Campaign analysis
 - Customer segmentation
 - SEO drafts
-

44. HR

Possible workflows:

```
Resume
↓
AI Extraction
↓
Structured Candidate Data
↓
Recruitment System
```

or:

```
Employee Question
↓
Company RAG System
↓
HR Policy Answer
```

45. Finance

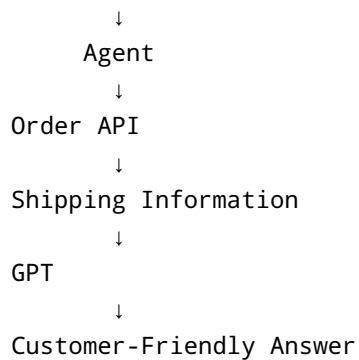
Examples:

```
Invoice
↓
AI Extraction
↓
Vendor
Amount
Date
Category
↓
Accounting Workflow
```

46. Ecommerce

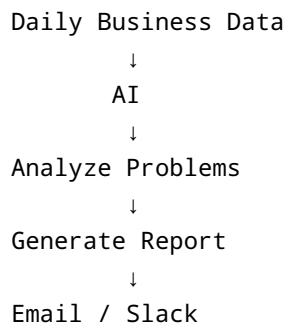
Example:

Customer:
"Where is order #12345?"



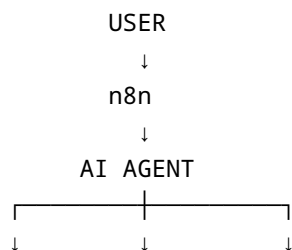
47. Operations

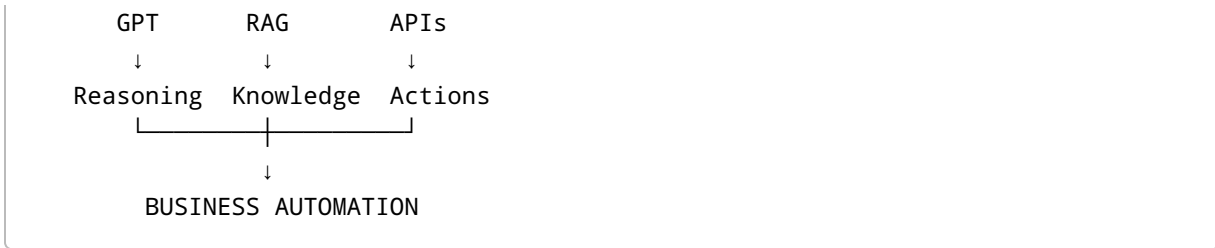
Example:



48. Connecting Day 1 to the Entire Course

Now show students the big architecture.





Explain each component.

GPT

The language intelligence.

AI Agent

Coordinates decisions and tools.

RAG

Provides relevant private or external knowledge.

API Integration

Connects AI with software.

n8n

Orchestrates workflows.

Business Automation

Uses everything together to solve a business problem.

49. Important Architecture Example

Suppose a customer asks:

Where is my order?

A basic GPT system might guess.

That is bad.

A professional system works like this:

```
Customer
↓
AI Agent
↓
Understand:
"Customer wants order status"
↓
Call Order Tool
↓
Shopify API
↓
Order:
#83421
Shipped
Tracking: XYZ
↓
Return Result to GPT
↓
GPT writes:
"Your order has shipped..."
```

Explain:

The **AI handles language and decision-making**.

The **API provides factual live data**.

50. RAG Example

Customer asks:

Can I return an opened product?

Architecture:

```
Question
↓
RAG Search
↓
Return Policy
↓
```

Relevant Section

↓

GPT

↓

Grounded Answer

The source of truth is the company policy.

51. Agent Example

Suppose an AI sales agent has these tools:

```
search_customer()  
create_lead()  
send_email()  
schedule_meeting()  
update_crm()
```

User says:

Follow up with John and schedule a demo if he is interested.

The agent may:

```
Find John  
↓  
Read CRM Data  
↓  
Draft Follow-Up  
↓  
Send Email  
↓  
Wait for Response  
↓  
Detect Interest  
↓  
Check Calendar  
↓  
Schedule Meeting  
↓  
Update CRM
```

Students do not need to build this today.

They only need to understand where the course is going.

52. Live Demonstration — Suggested 5 Minutes

Use a general-purpose AI assistant.

Demo 1 — Summarization

Provide:

```
Our company sells organic skincare products.  
Most customers are women aged 25-45.  
Our highest-selling product is Vitamin C Serum.  
Customer complaints have increased because deliveries are taking  
7-10 days instead of our advertised 3-5 days.
```

Prompt:

```
Summarize this business information in three bullet points.
```

Discuss the output.

53. Demo 2 — Classification

Prompt:

```
Analyze this customer message:  
  
"I've contacted you three times.  
My package still hasn't arrived.  
I need this fixed immediately."  
  
Return:  
Category  
Sentiment  
Priority  
Summary
```

Expected concept:

Category: Shipping Issue
Sentiment: Negative
Priority: High
Summary: Customer has repeatedly contacted support about a missing package.

54. Demo 3 — Extraction

Input:

Hello, my name is David Chen.
I need 30 office chairs delivered to Denver before September 15.
My budget is \$7,500.

Prompt:

Extract the important business information.

Possible output:

```
{  
  "customer": "David Chen",  
  "product": "office chairs",  
  "quantity": 30,  
  "location": "Denver",  
  "deadline": "September 15",  
  "budget": 7500  
}
```

Explain:

This ability is extremely important for automation.

55. Demo 4 — Generation

Prompt:

Write a professional reply confirming that our sales team will review David's request.

Students can observe the difference between:

Extraction

and

Generation.

56. Demo 5 — AI Limitations

Ask the AI:

How many office chairs does David's warehouse currently have?

Point out:

That information was never provided.

Ask students:

What should the correct AI response be?

Correct behavior:

I don't have enough information to determine current warehouse inventory.

Introduce the concept:

Don't invent missing information.

57. Student Exercise — 5 Minutes

Give students this customer message:

Hi, I ordered a laptop from your store nine days ago.
The website said delivery would take 3-5 days.
Tracking has not updated for four days.
This laptop is needed for work, so please tell me what is happening.

Ask students to use AI to produce:

1. Summary
2. Intent
3. Sentiment
4. Priority
5. Information required to solve the issue
6. Suggested response

Then ask:

What information does the AI NOT know?

Expected answers:

- Actual order number
- Current shipping status
- Courier information
- Customer identity
- Warehouse information

Ask:

How could AI obtain this information?

Expected answer:

API / business system integration.

This naturally prepares students for future lessons.

58. Day 1 Mini Discussion

Ask students:

Question 1

Can GPT generate text?

Answer:

Yes.

Question 2

Does GPT automatically know your company's private documents?

Answer:

No.

Question 3

Can GPT automatically know current inventory?

Answer:

No, unless the relevant information is provided through context or tools.

Question 4

Can GPT send an email by itself?

Answer:

A language model alone does not perform that external action. It requires a connected application/tool.

Question 5

What can provide private company knowledge?

Answer:

For example:

RAG / connected knowledge systems.

Question 6

What can provide live business information?

Answer:

APIs / databases / tools.

Question 7

What can coordinate actions?

Answer:

AI Agents and workflow automation.

59. Day 1 Quiz

Question 1

What does AI stand for?

- A. Automated Internet
- B. Artificial Intelligence
- C. Advanced Integration
- D. Automated Information

Answer: B

Question 2

What does GPT stand for?

- A. General Processing Technology
- B. Generative Pre-trained Transformer
- C. Generated Prompt Tool
- D. Global Prediction Technology

Answer: B

Question 3

What is an LLM?

- A. Large Language Model
- B. Language Learning Machine
- C. Logical Language Manager
- D. Large Learning Method

Answer: A

Question 4

Which is an example of Generative AI?

- A. Creating an email
- B. Detecting a keyboard press
- C. Turning a computer on
- D. Copying a file

Answer: A

Question 5

What is a token?

- A. A cryptocurrency only
- B. A unit used by language models to process text
- C. A database
- D. An API key

Answer: B

Question 6

What is a prompt?

- A. A database table
- B. An instruction/input given to AI
- C. A programming language
- D. A vector database

Answer: B

Question 7

What is context?

- A. Information available to the model for the current interaction
- B. Only the user's latest sentence
- C. A type of computer
- D. An API

Answer: A

Question 8

What is hallucination?

- A. AI generating unsupported or incorrect information
- B. AI processing an image
- C. AI calling an API
- D. AI storing information

Answer: A

Question 9

What is inference?

- A. Training an AI model from the beginning
- B. Using a trained model to generate an output
- C. Installing Python
- D. Creating a database

Answer: B

Question 10

Which architecture is more appropriate for current order information?

A.

Customer → GPT Guess

B.

Customer → GPT/Agent → Order API → Current Data

Answer: B

60. Key Vocabulary

Students should leave Day 1 knowing these terms.

Artificial Intelligence

Technology designed to perform tasks associated with intelligent behavior.

Machine Learning

A method where systems learn patterns from data.

Generative AI

AI capable of generating new content.

LLM

Large Language Model.

GPT

Generative Pre-trained Transformer.

Model

A learned computational system that maps inputs to outputs.

Prompt

Input or instructions sent to the AI.

Token

A unit used by language models to process text.

Context

Information available to the model during an interaction.

Context Window

The amount of information a model can work with at one time.

Training

The process used to teach a model patterns from data.

Inference

Using a trained model to generate an output.

Hallucination

An unsupported or incorrect AI-generated statement.

API

A mechanism that enables software systems to communicate.

RAG

Retrieval-Augmented Generation: retrieving relevant information and providing it to a generative model.

AI Agent

An AI-driven system that can make decisions and use tools to pursue a goal.

Automation

Using technology to execute work with reduced manual involvement.

61. Ten Important Day 1 Rules

Students should remember these ten ideas.

Rule 1

AI is broader than ChatGPT.

Rule 2

ChatGPT is an application; GPT is a model family.

Rule 3

LLMs generate language based on learned patterns and context.

Rule 4

A fluent AI response is not automatically correct.

Rule 5

AI needs relevant context to make better decisions.

Rule 6

Private company knowledge usually needs to be supplied or retrieved.

Rule 7

Live business information usually requires external systems or tools.

Rule 8

Actions require integrations.

Rule 9

High-risk actions may require human approval.

Rule 10

Professional AI solutions combine AI with data, tools, workflows, and business rules.

62. Homework Assignment

Task 1 — Find Five AI Opportunities

Choose one business.

Examples:

- Ecommerce store
- Restaurant
- Real-estate company
- Marketing agency
- Law office
- Recruitment company
- Logistics company
- Hotel
- Healthcare administration
- Software company

Identify five tasks AI could help with.

Use this format:

```
Business:
Ecommerce Store

Task 1:
Customer-support emails

Current Problem:
Employees manually classify hundreds of emails.

AI Solution:
GPT classifies each email by intent and priority.

Expected Benefit:
Reduce manual triage work.
```

Students should create five examples.

63. Homework Task 2 — AI vs Non-AI

Classify these.

```
Calculator
Spam Detector
ChatGPT
Automatic Door
Netflix Recommendation Engine
```

AI Image Generator
Excel Formula
Customer Support Agent
RAG Assistant
Traditional Website Contact Form

Students decide:

AI
or
Not necessarily AI

Then explain why.

64. Homework Task 3 — Use GPT for Five Different Tasks

Ask students to use an AI assistant for:

1. Summarization

Summarize one article or document.

2. Classification

Classify five customer messages.

3. Extraction

Extract names, dates, and numbers from a paragraph.

4. Generation

Generate one professional email.

5. Analysis

Analyze ten customer reviews and identify common complaints.

For every task, students should save:

Prompt
Output
What worked
What failed
What could be improved

65. Optional Challenge

Give students this scenario:

ABC Fashion receives 1,000 customer messages per day.

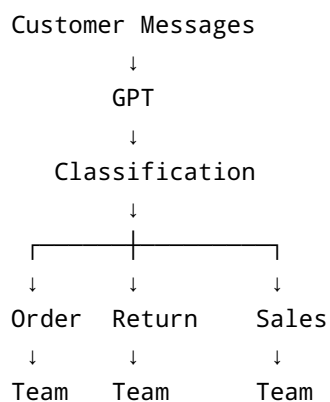
Messages include:

- Order status
- Returns
- Refunds
- Product questions
- Complaints
- Sales inquiries

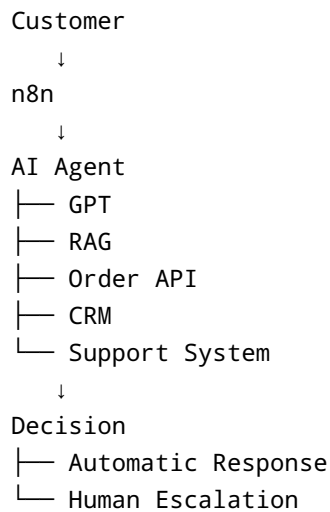
Ask:

How could AI help?

A beginner-level proposed architecture could be:



Advanced future architecture:



Tell students:

By Day 30, you should understand how to design the second architecture.

66. Instructor Closing Script

End the lecture with this message:

Today you learned the foundation of modern generative AI.

The most important concept is that **GPT is not an entire business automation system.**

GPT provides powerful language intelligence.

To build professional systems, we will gradually add:

```
GPT
+
Prompt Engineering
+
Structured Outputs
+
RAG
+
AI Agents
+
APIs
+
```

```
n8n
+
Business Rules
=
AI Business Automation
```

During the next 29 days, we will learn each layer and eventually combine them into complete AI systems.

67. Day 1 Final Summary

Students should now be able to explain:

What is AI?

A broad field of building systems that perform tasks associated with intelligence.

What is Generative AI?

AI that can create new content.

What is an LLM?

A Large Language Model.

What is GPT?

A Generative Pre-trained Transformer model family.

What is a prompt?

Instructions or information given to the AI.

What is a token?

A unit of text processed by the model.

What is context?

Information available to the model during generation.

What is training?

The process through which a model learns patterns.

What is inference?

Using a trained model to generate outputs.

What is hallucination?

AI-generated information that is unsupported or incorrect.

What does a business AI system require beyond GPT?

Potentially:

```
Knowledge
+
Tools
+
APIs
+
Agents
+
Workflow Automation
+
Business Rules
+
Human Oversight
```

Day 1 Learning Outcome

At the end of the 60-minute lecture, a student should be able to look at:

```
Customer Question
```

and understand that there is a major difference between:

```
Customer
↓
GPT
↓
Generated Answer
```

and a professional system:



That mental model is the **most important objective of Day 1**.